

Paper presented at the International Rice Congress 2023 from October 16-19 in Manila, Philippines.

Method Validation of a Detection Protocol of *Xanthomonas oryzae* pv. *oryzae* and *X. o.* pv. *oryzicola* in Rice Seeds

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Bacterial leaf blight (BLB) and bacterial leaf streak (BLS) caused by *Xanthomonas oryzae* pv. *oryzae* (Xoo) and *Xanthomonas oryzae* pv. *oryzicola* (Xoc), respectively, are the two most important bacterial diseases of rice worldwide. Xoo and Xoc are widely distributed and endemic in many rice producing countries throughout Asia, Africa and Australia. The expansion of international trade in agricultural seeds and products poses challenges in effectively managing the spread and movement of plant pathogens into new agroecosystems. Rapid detection and accurate identification of pathogens are critical steps in effectively implementing pathogen containment or elimination responses. There are several diagnostic tools currently available for the identification of Xoo and Xoc. A multiplex polymerase chain reaction (PCR) technique utilizing pathovar-specific primers has been developed based on a comparative analysis of genomes of Xoo and Xoc (Lang et al., 2010). This method is widely used and is being employed for detecting and diagnosing Xoo and Xoc in rice seeds. The project will involve method validation of a qualitative seed testing protocol with rice seed extract using the multiplex PCR assay with the pathovar specific primers (Vera Cruz et al., 2013; Lang et al., 2010). The protocol includes two main parts - (i) sample preparation using sonication, and (ii) direct PCR assay with multiplex primers. The objective of this project is to reintroduce an earlier initiative of validating the protocol through the participation of many laboratories, in accordance with the method validation process established by the International Seed Testing Association (ISTA). Primarily, to conduct a comprehensive analysis of the performance criteria and present well documented proof of method performance, with the intention of emphasizing the method's reliability. The method validation will be participated by laboratories and National Plant Protection Organization laboratories and partner laboratories from Philippines and Africa; and jointly organized by IRRI Seed Health and Logistics Unit, AfricaRice Center and Bureau of Plant Industry-Central Post-Entry Quarantine Station.